

SAVITRIBAI PHULE PUNE UNIVERSITY

Digital Forensics II

CDS-361 · Complete Exam Study Guide

TY B.Sc. (Cyber and Digital Science) · Semester VI · 2020 Pattern

Exam Duration	3 Hours
Maximum Marks (UA)	70 Marks
Continuous Assessment	30 Marks
Total Questions	Q1–Q5 (All Compulsory)
Content Coverage	Previous Year PYQs + Predicted Questions
Chapters Covered	Email Mobile Multimedia Cloud VM & Network Forensics

● Email & Social
Media● Mobile
Forensics● Multimedia
Forensics

● Cloud Forensics

● VM & Network

■ Predicted

Q1 — Attempt Any TEN of the following [10 × 1 = 10 Marks]**Q1(a) What are Windows Prefetch Artifacts?****[1 Mark]****● Cloud / Windows**

Windows Prefetch Artifacts are files stored in C:\Windows\Prefetch\ with extension .pf, automatically created by Windows to speed up application loading. Each .pf file records: the application name, number of times executed, last run timestamp, and a list of files/directories accessed during execution.

Forensic Value: Proves a program was executed on the system, shows execution count and last-run timestamp — critical evidence even if the original executable has been deleted.

Q1(b) Define Vector Graphics.**[1 Mark]****● Multimedia Forensics**

Vector graphics are images defined using mathematical formulas — points, lines, curves, and geometric shapes — rather than a grid of pixels. They are infinitely scalable without any loss of quality or pixelation.

Examples: SVG, AI (Adobe Illustrator), EPS, CDR files. **Key property:** Resolution-independent.

Q1(c) Define Service Level Agreements (SLA).**[1 Mark]****● Cloud Forensics**

An SLA (Service Level Agreement) is a formal contract between a Cloud Service Provider (CSP) and customer defining the expected level of service — including uptime guarantees, data access rights, security responsibilities, log retention periods, and breach notification obligations.

Forensic relevance: SLAs define what forensic data the CSP must provide investigators, how long logs are retained, and jurisdictional constraints on evidence access.

Q1(d) What is file carving?**[1 Mark]****● Mobile / General Forensics**

File carving is a data recovery technique that extracts files from raw disk or memory images by identifying known file signatures (magic bytes/headers) and footers, without relying on any file system metadata or directory structures.

Used when: File system is corrupted, files are deleted, or for unallocated space recovery. **Tools:** Foremost, Scalpel, Autopsy.

Q1(e) Define Metafile Graphic.**[1 Mark]****● Multimedia Forensics**

A Metafile graphic is a file format that can contain both vector (drawing commands/instructions) and raster (bitmap pixel) graphic data together within a single file. The image is rendered by replaying stored

drawing commands.

Examples: WMF (Windows Metafile), EMF (Enhanced Metafile), PICT.

Q1(f) What is Steganography?

[1
Mark]

● **Multimedia Forensics**

Steganography is the practice of concealing secret information within an ordinary, non-secret carrier file (image, audio, video, or text) without altering its visible or audible appearance, so the existence of the hidden message is not detectable.

Common technique: LSB (Least Significant Bit) substitution — replacing last bit of each pixel color value with a secret data bit. **Detection:** Steganalysis tools like StegDetect, OpenStego.

Q1(g) What is logical data extraction?

[1
Mark]

● **Mobile Forensics**

Logical data extraction is a mobile forensics acquisition method that uses the device's OS and standard protocols (e.g., ADB for Android) to extract an organized copy of accessible files — contacts, messages, call logs, app data — without accessing raw storage or flash memory directly.

Limitation: Cannot recover deleted data. Only retrieves files accessible via OS APIs. **Tool:** adb pull, Oxygen Forensics, Cellebrite UFED.

Q1(h) What is email investigation?

[1
Mark]

● **Email & Social Media**

Email investigation is the forensic process of examining email messages and their associated metadata to trace the origin, transmission path, authenticity, and content of emails as part of a cybercrime investigation — such as fraud, harassment, phishing, or corporate espionage.

Key activities: Examining email headers, tracing sender IP, analyzing attachments, reviewing server logs, identifying spoofing/phishing indicators.

Q1(i) List some forensic tools used for social media investigations.

[1
Mark]

● **Email & Social Media**

- **Social Mapper** — cross-platform facial recognition to correlate profiles
- **Maltego** — link analysis and OSINT graph mapping
- **X1 Social Discovery** — collects and preserves social media evidence legally
- **Sherlock** — username hunting across 300+ social networks
- **Twint** — Twitter scraping without API authentication
- **OSINT Framework** — aggregates OSINT tools by category

Q1(j) Give one example of a Type 2 Hypervisor.

[1
Mark]

● **VM & Network Forensics**

Example: VMware Workstation (also accepted: Oracle VirtualBox, VMware Fusion, Parallels Desktop, QEMU/KVM in hosted mode).

A Type 2 Hypervisor (hosted hypervisor) runs as an application on top of a conventional host operating system. The host OS manages hardware resources; the hypervisor creates and manages virtual machines above it.

Contrast with Type 1: Type 1 (bare-metal) runs directly on hardware — e.g., VMware ESXi, Microsoft Hyper-V, Xen.

Q1(k) Define Virtual Network.

**[1
Mark]**

● **VM & Network Forensics**

A Virtual Network is a software-defined network infrastructure that allows virtual machines and virtual devices to communicate as if connected by physical network hardware — without requiring dedicated physical cables, switches, or routers.

VMware network types: Bridged (shares physical NIC), NAT (shared IP via host), Host-only (isolated from external networks).

Q1(l) Enlist any two raster graphic file formats.

**[1
Mark]**

● **Multimedia Forensics**

- 1. JPEG (.jpg/.jpeg)** — Joint Photographic Experts Group; lossy compression using DCT; best for photographs; does not support transparency
- 2. PNG (.png)** — Portable Network Graphics; lossless compression; supports transparency (alpha channel); best for screenshots, graphics with text, logos

Other valid answers: BMP, GIF, TIFF, RAW, WebP

Q2 — Attempt Any FIVE of the following [5 × 2 = 10 Marks]**Q2(a) Describe the process of extracting evidence from a mobile phone.****[2 Marks]****● Mobile Forensics**

- 1. Seizure & Isolation:** Place device in Faraday bag immediately to block all network signals — prevents remote wipe, data modification, or incoming calls that might overwrite call logs
- 2. Documentation:** Photograph device's physical state, note IMEI number, SIM card details, network carrier, battery level, and any damage before touching it
- 3. Preservation:** Charge device if needed; prevent it from locking (airplane mode); enable USB debugging via Developer Options if accessible
- 4. Acquisition:** Choose appropriate method — Manual (photography), Logical (ADB/Oxygen), File-system, or Physical (bit-by-bit) — based on device state and investigation needs
- 5. Hash Verification:** Compute MD5 and SHA-256 hash of the acquired image to establish and verify forensic integrity
- 6. Analysis:** Use Autopsy or Cellebrite to parse extracted data; recover deleted files via SQLite database parsing and file carving techniques
- 7. Reporting:** Document all tools used, commands executed, findings, chain of custody, and timestamps

Q2(b) Explain the differences between bitmap and raster images.**[2 Marks]****● Multimedia Forensics**

■ **Exam Trap** BMP (Bitmap) and Raster are often used interchangeably, but BMP is a specific file format while 'raster' is a broader image category. Always clarify this distinction.

Aspect	Bitmap (BMP format)	Raster (General Category)
Definition	Specific Microsoft file format storing pixel data with no/minimal compression	Any pixel-based image format (JPEG, PNG, GIF, TIFF, BMP)
Compression	Uncompressed or minimal (RLE only)	Varies: Lossless (PNG, TIFF) or Lossy (JPEG)
File size	Very large — no compression	Varies based on format and compression
Scope	Narrow — one specific format only	Broad — entire pixel-based image category
Scalability	Loses quality when enlarged (pixelation)	All raster images lose quality when enlarged

Q2(c) Why is examining email headers important in email forensics?**[2 Marks]****● Email & Social Media**

Email headers contain technical metadata added by each mail server as the message travels from sender to recipient. Key reasons for examining them:

→ **Tracing origin IP:** 'Received:' fields (read bottom-up) show each server hop; the lowest 'Received' entry reveals the actual sender's IP address

- **Detecting spoofing:** Comparing 'From:' vs 'Return-Path:' vs actual sending IP exposes forged sender addresses
- **Timestamp verification:** Each 'Received:' hop has timestamps that can reveal forged or manipulated send times
- **Authentication results:** DKIM-Signature, SPF result, DMARC policy indicate whether email is genuinely from claimed sender
- **Message-ID tracking:** Unique identifier used to correlate emails across multiple server logs

Q2(d) Name two tools used for cloud forensics.

[2 Marks]

● **Cloud Forensics**

1. **WinHex:** Hex editor and disk analysis tool used to examine cloud-synchronized files stored locally (Google Drive cache, OneDrive temp files, Dropbox metadata). Can recover deleted cloud artifacts from local disk prefetch data and registry entries.
2. **Elcomsoft Cloud Explorer / Magnet AXIOM:** Extracts data from cloud service accounts (iCloud, Google Account, Microsoft 365) including backups, photos, notes, and location history using device authentication tokens captured during forensic acquisition.

Also acceptable: Cellebrite UFED (cloud extraction), AWS CloudTrail (log analysis), FROST.

Q2(e) What is the primary goal of securing a network?

[2 Marks]

● **Network Forensics**

The primary goal of network security is to maintain the **CIA Triad** across all network resources:

- **Confidentiality:** Only authorized parties can access data; enforced via encryption, access control lists (ACLs), VPNs
- **Integrity:** Data is not altered in transit or storage; enforced via hashing (MD5/SHA), digital signatures, checksums
- **Availability:** Services remain accessible to authorized users; ensured through redundancy, load balancing, DDoS protection

Additional goals: **Authentication** (verifying identity), **Non-repudiation** (preventing denial of actions), and **Accountability** (tracing actions to individuals via logs).

Q2(f) List any two key responsibilities of Incident First Responders in cloud forensics.

[2 Marks]

● **Cloud Forensics**

1. **Evidence Preservation:** Immediately snapshot running virtual machine instances, capture volatile memory (RAM), and preserve cloud storage objects before auto-scaling, auto-deletion policies, or log rotation destroys critical evidence. Compute and record cryptographic hashes (MD5/SHA-256) of all captured data to prove integrity.
2. **Log Collection & Chain of Custody:** Collect all available logs from CSP — CloudTrail/Audit logs, IAM access logs, network flow logs, API call logs. Document timestamps and hash values of each log file. Coordinate with CSP legal/security team regarding jurisdiction, warrant compliance, and formal chain of custody documentation required for court admissibility.

Q3 — Attempt Any FOUR of the following [4 × 5 = 20 Marks]**Q3(a)****What factors should be considered while investigating Cloud Service Providers (CSPs)?****[5 Marks]**● **Cloud Forensics****1. Jurisdiction and Legal Authority**

Data may be physically stored in data centers across multiple countries. Investigators must determine: which country's laws apply, whether a court order is internationally valid, and if Mutual Legal Assistance Treaties (MLATs) are required. The SLA must be reviewed for law enforcement cooperation clauses. Without correct legal authority, evidence may be inadmissible.

2. Service Level Agreements (SLA)

The SLA defines: what forensic data the CSP is obligated to provide investigators, log retention periods (typically 30–90 days, after which data may be overwritten), the formal process for evidence/legal data requests, data encryption specifications, and customer data access procedures.

3. Multi-Tenancy Issues

Cloud environments share physical infrastructure across multiple customers. Acquiring a complete disk image of a cloud server would compromise data belonging to other tenants — creating serious legal liability. Investigators must use targeted, scoped acquisition methods rather than full disk imaging.

4. Log Availability and Integrity

Verify what logs exist and their retention status: API call logs (e.g., AWS CloudTrail), network flow logs, access/authentication logs, object storage logs. Confirm the CSP has not modified logs and obtain cryptographic proof of log integrity (hash certificates) if available.

5. Encryption and Key Management

Cloud data is often encrypted at rest and in transit. Investigators need decryption keys to access data. Keys may be held by the CSP, the customer (BYOK — Bring Your Own Key), or a third-party KMS. Determine key custodianship and obtain lawful access through proper channels.

6. Chain of Custody

Document precisely how evidence is obtained from the CSP, who accessed it, timestamps, and delivery method. CSP-provided data must be accompanied by a signed affidavit attesting to its authenticity and completeness for court admissibility.

KEY POINT

CSP investigation requires balancing forensic needs against multi-tenancy privacy, jurisdictional constraints, and SLA limitations — technical and legal factors are equally critical.

Q3(b)**Explain the difference between Lossless and Lossy Compression.****[5 Marks]**● **Multimedia Forensics**

Parameter	Lossless Compression	Lossy Compression
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Definition	Compresses data so original can be perfectly reconstructed without any data loss	Permanently removes some data (usually imperceptible) to achieve higher compression ratios
Data integrity	100% — original data fully preserved	Partial loss — some information permanently discarded
Compression ratio	Moderate (2:1 to 5:1 typically)	High (10:1 to 100:1 possible for images)
File size	Larger than lossy	Much smaller
Use case	Medical imaging, legal documents, archiving, text files, PNG images	Web photos, streaming video/audio, JPEG images, MP3 audio
Algorithms	ZIP (DEFLATE), Huffman coding, Run-Length Encoding (RLE), LZW	DCT (JPEG), Wavelet (JPEG2000), Perceptual coding (MP3, AAC)
Quality after multiple saves	No degradation regardless of saves	Progressive quality loss with each save/re-encode cycle
Forensic relevance	Preferred for evidence — preserves all pixel data for steganalysis	Lossy artifacts can complicate steganography detection

Forensic impact: Lossy compression (JPEG) can destroy steganographic content embedded in an image. Always work with lossless originals when conducting multimedia forensics and steganalysis.

Q3(c)

What are Type 2 Hypervisors, and how do they differ from Type 1 Hypervisors?

[5 Marks]

● **VM & Network Forensics**

A Type 2 Hypervisor (also called a 'hosted hypervisor') is virtualization software that runs as an application on top of an existing host operating system. The host OS manages hardware resources and the hypervisor relies on the host OS for device drivers and hardware access.

Examples of Type 2: VMware Workstation, Oracle VirtualBox, VMware Fusion (macOS), Parallels Desktop, QEMU (user mode).

Aspect	Type 1 (Bare-Metal)	Type 2 (Hosted)
Runs on	Directly on hardware — no host OS	On top of a host operating system
Performance	High — direct hardware access	Lower — extra OS layer adds overhead
Examples	VMware ESXi, Hyper-V, Xen, KVM	VMware Workstation, VirtualBox, Parallels
Use case	Enterprise servers, data centers, cloud infrastructure	Developer workstations, testing, forensic labs
Resource management	Full control over hardware resources	Shares resources with host OS
Security isolation	Stronger — no host OS attack surface	Weaker — host OS vulnerabilities affect VMs
Forensic use	Cloud infrastructure examination	Creating isolated analysis environments

Forensic relevance of Type 2: Investigators use Type 2 hypervisors (VMware Workstation) to safely analyze malware in isolated VMs, examine forensic images, and conduct investigations without contaminating the host system. VM files (.vmdk, .vmx, .vmem) are themselves valuable artifacts.

Q3(d) Explain different Android data extraction techniques with examples.**[5 Marks]****● Mobile Forensics****1. Manual Data Extraction**

Investigator physically operates the device — navigating menus, photographing the screen, manually browsing contacts, messages, settings. No special tools needed but only visible/accessible data is captured. Slowest method; risk of contamination. Used as initial triage or when other methods are unavailable.

2. Logical Data Extraction

Uses Android Debug Bridge (ADB) or forensic tools to extract a logical copy of accessible files via the OS. Commands: 'adb pull /sdcard/' extracts SD card contents. SQLite Browser used to view extracted databases (call logs: /data/data/com.android.providers.contacts/databases/, SMS: /data/data/com.android.providers.telephony/). Cannot recover deleted data. Tools: Oxygen Forensics, Cellebrite UFED.

3. Physical Data Extraction

Creates a bit-by-bit (sector-by-sector) image of the entire device storage including deleted data, unallocated space, and app databases. Requires device to be rooted or exploiting a vulnerability. Methods: 'adb shell dd if=/dev/block/mmcblk0 | adb pull' or JTAG/Chip-off for locked/damaged devices. Most comprehensive — allows file carving for deleted recovery.

4. File-System Acquisition

Acquires a copy of the device's file system (rather than raw sector image) including deleted file entries. Intermediate between logical and physical. Requires root access. Tools: Cellebrite UFED, MSAB XRY.

5. JTAG and Chip-Off Acquisition

JTAG: Connects to the device's debug port (JTAG interface) to read raw memory. Used when device is locked, damaged, or cannot boot. **Chip-Off:** Physically removes the flash memory chip from the PCB and reads it directly using specialized readers. Last resort for severely damaged devices. Both methods require specialist hardware skills.

Q3(e) Which are two methods of email message tracing?**[5 Marks]****● Email & Social Media****Method 1: Email Header Analysis**

Every email carries a chain of 'Received:' headers added by each mail server that handled the message. By reading these headers from bottom (origin) to top (destination), investigators can trace the complete path of an email through the internet.

- The bottom-most 'Received:' header contains the originating IP address of the sender
- Each header includes: server hostname, IP address, timestamp, and protocol used
- Cross-reference the originating IP with WHOIS / geolocation databases to identify ISP and approximate location
- Verify DKIM signatures and SPF records to confirm or deny sender authentication

→ Tool: MX Toolbox Email Header Analyzer, Google Admin Toolbox, manual inspection

Method 2: Server Log Analysis / Network Email Log Tracing

Mail server logs (SMTP logs) record every email transaction — sender, recipient, source IP, timestamp, message-ID, and delivery status. Investigators can request these logs from the email service provider under a lawful court order.

- **Unix/Sendmail/Postfix logs:** Located at /var/log/mail.log — contain full SMTP transaction records
- **Microsoft Exchange logs:** Message Tracking Logs show sender/recipient details, server hops, and delivery status
- **Gmail/Office365:** Admin console provides message header analysis and delivery reports
- Message-ID field (unique identifier) can be cross-referenced across multiple server logs to track the exact path
- IP address correlation across logs can identify the true sender even if the From: field was spoofed

EXAM TIP

Always mention BOTH technical tracing (header analysis) AND log-based tracing for full marks. Emphasize that headers can be forged but server logs are generally more reliable for forensic purposes.

Q4 — Attempt Any FOUR of the following [4 × 5 = 20 Marks]

Q4(a) What is live acquisition? Why is it important in digital forensics?

[5 Marks]

● VM & Network Forensics

Live acquisition (also called 'live forensics') is the process of collecting digital evidence from a running (powered-on) system without shutting it down first — capturing volatile data that exists only while the system is active.

What is captured during live acquisition?

- **RAM contents:** Running processes, encryption keys, passwords in memory, open network connections, clipboard data
- **Network state:** Current connections (netstat), ARP cache, routing tables, open sockets
- **Running processes:** Process list, loaded DLLs, process memory dumps
- **Logged-in users:** Active user sessions, recently accessed files
- **System time and timezone:** Establishes forensic timeline

Why is live acquisition important?

- **Volatile data loss:** RAM contents are permanently lost when the system is powered off — encryption keys for BitLocker/VeraCrypt exist only in RAM
- **Encryption:** Modern full-disk encryption (FDE) means powered-off systems present encrypted, unreadable disks — live acquisition captures data while decrypted
- **Malware in memory:** Fileless malware (living-in-memory) leaves no disk artifacts — only RAM acquisition can capture it
- **Network forensics:** Active connections may reveal C2 (Command & Control) servers; connections terminate when system shuts down
- **Order of Volatility:** RFC 3227 mandates collecting most volatile evidence first — live acquisition follows this order: RAM → Swap → Network → Running processes → Disk

Windows Live Acquisition steps (using Sysinternals/FTK)

1. Use write-protected USB with forensic tools; run as Administrator
2. Capture running process list: tasklist /v > processes.txt
3. Capture network connections: netstat -ano > netstat.txt
4. Capture RAM using DumpIt.exe or FTK Imager (File > Capture Memory)
5. Hash all captured files (md5sum/sha256sum)
6. Document all actions with timestamps

Q4(b) What are the basic concepts of cloud forensics?

[5 Marks]

● Cloud Forensics

1. Cloud Computing Overview

Cloud computing provides on-demand access to shared computing resources (servers, storage, networking, software) over the internet. Cloud forensics is the application of digital forensic principles to data and services hosted in cloud environments.

2. Cloud Service Models

Model	What Provider Manages	Forensic Challenge
IaaS (Infrastructure as a Service)	Hardware, networking, storage	Customer controls OS — forensic access to VMs possible
PaaS (Platform as a Service)	OS, middleware, runtime	Limited access — provider controls infrastructure logs
SaaS (Software as a Service)	Everything including application	Minimal access — only application-level logs available

3. Cloud Deployment Models

- **Public Cloud** (AWS, Azure, GCP) — shared infrastructure; multi-tenancy issues
- **Private Cloud** — single organization; easier forensic access
- **Hybrid Cloud** — mix of public/private; complex evidence collection
- **Community Cloud** — shared by specific industry group

4. Core Forensic Challenges

- **Jurisdiction:** Data distributed across global data centers — conflicting national laws
- **Multi-tenancy:** Cannot image shared physical hardware without affecting other customers
- **Volatile infrastructure:** VMs may be auto-deleted; evidence destroyed by auto-scaling
- **Log limitations:** CSPs control what logs are generated and retained
- **Encryption:** Data encrypted at rest/in transit — key access is a legal challenge

5. Evidence Sources in Cloud

- VM disk images and memory snapshots
- CSP audit logs (AWS CloudTrail, Azure Monitor)
- Access logs, API call logs, IAM logs
- Network flow logs (VPC Flow Logs)
- Local artifacts on client machines (Prefetch files, registry entries for cloud apps)

Q4(c)

How does the SQLite database help in recovering deleted data from mobile devices?

[5 Marks]

● Mobile Forensics

SQLite is a lightweight, self-contained relational database engine used extensively by Android and iOS applications to store structured data — contacts, call logs, SMS/MMS, browser history, social media chats, and app-specific data — in .db files on device storage.

SQLite Database Architecture and Recovery

SQLite stores data in fixed-size pages (default 4096 bytes). When a record is deleted:

- The record's page is marked as 'free' in the database freelist but not immediately overwritten
- The deleted record's data remains physically present in the database file until the page is reused
- SQLite's freelist pages and unallocated space within pages contain recoverable deleted records
- The database's Write-Ahead Log (WAL) file may also contain recent transactions including deleted data

Recovery Techniques

- 1. SQLite Browser / DB Browser:** Open .db file and browse tables — visually inspect records and deleted entries in freelist
- 2. Forensic carving of SQLite pages:** Tools like 'sqlparse' and 'undark' parse the raw binary of .db files to recover deleted records from unallocated pages
- 3. WAL file analysis:** Parse the Write-Ahead Log (.db-wal) file which contains uncommitted/recent transactions
- 4. File carving on disk image:** Even if .db file is deleted, use file signatures (SQLite magic: 0x53514c69...) to carve entire database from unallocated space
- 5. Autopsy plugin:** Android Analyzer module automatically parses and presents SQLite databases with deleted record recovery

Common SQLite databases on Android

- contacts2.db — contacts and call logs (com.android.providers.contacts)
- mmssms.db — SMS/MMS messages (com.android.providers.telephony)
- browser2.db — browser history (com.android.browser)
- WhatsApp: msgstore.db — all WhatsApp messages
- Instagram: direct.db — direct messages

EXAM TIP

Key point: SQLite does NOT immediately overwrite deleted records — pages are marked free but data persists until reused. This is why SQLite is such a rich source of recoverable evidence.

Q4(d) Explain the role of email headers in investigating email crimes.

[5 Marks]

● Email & Social Media

Email headers are structured metadata fields that document the technical journey of an email from sender to recipient. Every Mail Transfer Agent (MTA) that handles a message appends a 'Received:' header. These headers are the primary forensic artifact for email crime investigation.

Key Email Header Fields and Their Forensic Role

Header Field	Content	Forensic Use
Received:	Server hostnames, IPs, timestamps for each hop	Reconstruct full email path; find originating IP
From:	Claimed sender's display name and address	Can be forged — must cross-check with 'Return-Path'
Return-Path:	Where bounces are sent — usually real sender address	More reliable than 'From:' for identifying sender
Message-ID:	Globally unique identifier for the message	Cross-reference across server logs to track email
DKIM-Signature:	Cryptographic signature by sending domain	Verifies email was not modified after sending
X-Originating-IP:	Webmail clients add this — actual sender IP	Critical for Gmail/Yahoo investigations

Authentication-Results:	SPF/DKIM/DMARC pass/fail results	Determines if sender domain is verified
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Investigation Process

1. Extract full headers (Gmail: 'Show Original'; Outlook: File > Properties)
2. Read 'Received:' chain from bottom to top to trace message path
3. Identify originating IP address from lowest 'Received:' header
4. WHOIS/GeoIP lookup on originating IP to find ISP and location
5. Subpoena ISP for account records associated with that IP at that timestamp
6. Verify DKIM/SPF/DMARC to confirm or deny domain authenticity
7. Cross-reference Message-ID with mail server logs for corroboration

Q4(e) Describe the use of the Steganalysis Tool.

[5 Marks]

● Multimedia Forensics

Steganalysis is the practice of detecting, identifying, and extracting hidden information that has been concealed within digital media files (images, audio, video, text) using steganography techniques. Steganalysis tools automate this detection and extraction process.

Types of Steganalysis Techniques

- **Visual Steganalysis:** Visually inspect the carrier file for anomalies — unusual pixel patterns, color distortions, or visible artifacts introduced by data hiding
- **Statistical Steganalysis:** Analyze statistical properties of the file. Steganography alters natural statistical distributions of pixel values or frequency domain coefficients. Chi-square test detects LSB steganography
- **Signature/Pattern Steganalysis:** Different steganography tools leave unique signatures in carrier files — database of known tool signatures can detect which tool was used
- **Structural Steganalysis:** Examines file structure/metadata anomalies — unusual file size (larger than expected for content), modified EXIF data, non-standard compression artifacts

Common Steganalysis Tools

- **StegDetect:** Detects steganography created by JSteg, JPHide, Invisible Secrets, OutGuess, F5 in JPEG images using statistical analysis
- **StegExpose:** Command-line batch steganalysis combining multiple detection algorithms; gives confidence scores
- **OpenStego:** Can both hide and detect hidden data; useful for testing known algorithms
- **SteghideExtract:** Extracts hidden data from files created with Steghide tool if passphrase is known
- **Autopsy Steganalysis plugin:** Integrated into forensic workflow — flags suspicious images for further analysis
- **Stegsolve:** Visual analysis tool — applies color filters, plane analysis, and bit plane manipulation to reveal hidden content

Forensic Process using Steganalysis

1. Identify suspicious files (unusually large images, mismatched file metadata)
2. Run statistical analysis (chi-square test) to detect LSB embedding patterns
3. Apply signature-based detection to identify which steganography tool was used

4. If tool identified, use corresponding extraction tool with recovered passphrase
5. Document findings: carrier file hash, tool used, extracted content, method

Q5 — Attempt Any ONE of the following [1 × 10 = 10 Marks]**Q5 — OPTION A****Q5(a)
[5M]****Explore the role of email investigation.****[5
Marks]****● Email & Social Media**

Email investigation plays a central role in modern cybercrime detection and prosecution. Emails are involved in phishing attacks, corporate espionage, fraud, harassment, and insider threats. The forensic examination of email systems provides both direct evidence (content) and metadata evidence (headers, logs).

Role 1: Establishing Communication Records

Email logs and messages establish a documented record of communication between parties — crucial for proving conspiracy, coordination of criminal activity, or evidence of intent. In corporate investigations, email trails reveal unauthorized data exfiltration or policy violations.

Role 2: Identifying and Tracing Perpetrators

Through header analysis, investigators can trace the originating IP address of any email. Even with spoofed addresses, the X-Originating-IP or the bottom-most 'Received:' header typically reveals the true source. Combined with ISP records, this identifies the sender. DKIM signatures confirm if an email genuinely originated from the claimed domain.

Role 3: Detecting Phishing and Social Engineering Attacks

Email forensics identifies phishing campaigns by analyzing: spoofed domain names (typosquatting), fraudulent hyperlinks in HTML emails, malicious attachments (analyzing MIME types and encoded content), and sender reputation (checking IP blacklists, SPF failures). This evidence is used both to prosecute attackers and to improve organizational defenses.

Role 4: Evidence in Legal Proceedings

Email evidence is admissible in court when properly collected with maintained chain of custody. Email headers provide timestamped, server-corroborated records that are difficult to fabricate. Message-IDs allow cross-referencing across multiple server logs to prove authenticity.

Role 5: Corporate Policy Enforcement

Email investigation helps organizations enforce acceptable use policies by: detecting data leakage (sensitive documents emailed externally), identifying insider threats (employees communicating with competitors), monitoring for harassment or discrimination, and ensuring regulatory compliance (GDPR, HIPAA — verifying sensitive data handling).

**Q5(b)
[3M]****Which tools are commonly used for viewing forensic images?****[3
Marks]****● General Forensics**

→ **FTK Imager (AccessData):** Industry-standard tool for creating and viewing forensic images in E01, DD, AFF, S01 formats. Provides directory tree view, hex viewer, and file preview. Free to download.

- **Autopsy:** Open-source digital forensics platform. Views forensic images and provides full investigation capabilities — file carving, keyword search, timeline analysis, email parsing, web artifact analysis.
- **EnCase Investigator:** Commercial enterprise forensics tool from OpenText. Views LEF/E01 images with comprehensive analysis capabilities. Industry standard for law enforcement.
- **WinHex:** Hex editor with forensic image support. Views raw disk images at sector level. Used for manual artifact examination and data carving.
- **Volatility:** For memory (RAM) forensic images — analyzes process lists, network connections, registry hives from memory dumps.

Q5(c)
[2M]

Differentiate between email client and email server.

[2 Marks]

● **Email & Social Media**

Aspect	Email Client	Email Server
Definition	Software application used by end-users to compose, send, receive, and manage emails	Dedicated server software that routes, stores, delivers, and manages email traffic between users
Examples	Microsoft Outlook, Mozilla Thunderbird, Gmail (webmail), Apple Mail	Microsoft Exchange, Postfix, Sendmail, Dovecot, Gmail backend (Google)
Protocols used	Sends: SMTP (port 587/465); Receives: IMAP (port 993) or POP3 (port 995)	MTA uses SMTP (port 25) for server-to-server transfer; MDA handles local delivery
Data location	Local machine (PST/OST/MBOX files) or IMAP server sync	Centralized storage — all user mailboxes stored on server
Forensic artifacts	PST/OST files, MBOX, email caches, attachments in Temp folders	Mail logs (/var/log/mail.log), message stores, delivery logs, user account records

Q5 — OPTION B (OR)

Q5(a)
[5M]

What is the need for Mobile Forensics? List down challenges in Mobile Forensics.

[5 Marks]

● **Mobile Forensics**

Need for Mobile Forensics

Mobile devices have become the primary computing platform for billions of people worldwide. They contain extraordinarily rich repositories of personal and behavioral data — making them critical evidence sources in virtually every type of criminal investigation.

- **Ubiquity of evidence:** Modern smartphones contain GPS location history, communications (calls, SMS, WhatsApp, email), social media activity, photographs with EXIF metadata, financial transactions, and biometric data
- **Crime tool and target:** Mobile devices are used to commit cybercrimes (phishing, banking fraud, CSAM distribution) and are targets of theft, hacking, and stalking
- **Corporate investigations:** Employee devices may contain evidence of data theft, policy violations, or corporate espionage

- **Counter-terrorism:** Terrorist networks use encrypted messaging on mobile devices — forensic extraction is critical for law enforcement
- **Court evidence:** Mobile forensics produces legally admissible digital evidence for criminal prosecutions

Challenges in Mobile Forensics

- **Device diversity:** Thousands of different device models, manufacturers (Samsung, Huawei, Apple, Xiaomi), and hardware configurations — no single tool works on all devices
- **OS fragmentation:** Multiple Android versions (5.0–14), iOS versions, and custom manufacturer ROMs (Samsung OneUI, MIUI) each with different security models and file system structures
- **Encryption:** Full-disk encryption (FDE) on Android 6+, File-Based Encryption (FBE) on Android 9+ — powered-off devices present encrypted, unreadable storage
- **Screen locks and biometrics:** PINs, passwords, patterns, face unlock, fingerprint — newer secure enclaves (Titan M, Secure Enclave) make brute-force practically impossible
- **Remote wipe:** Devices can be remotely wiped via Find My iPhone / Google Find My Device — requires immediate Faraday isolation to prevent signal reception
- **Cloud sync:** Evidence may be stored in cloud accounts (iCloud, Google Account) rather than locally on device — requires separate legal process to access
- **Anti-forensics:** Users may enable encryption, use privacy-focused apps (Signal with disappearing messages), or root device to remove forensic artifacts
- **Rapid technology evolution:** New SoC architectures, memory types (UFS 3.1), and OS security features regularly outpace forensic tool capabilities

Q5(b)
[3M]

List some major cloud vendors and their key features.

[3 Marks]

● Cloud Forensics

Vendor	Key Features	Forensic Relevance
Amazon Web Services (AWS)	EC2 (VMs), S3 (storage), Lambda (serverless), CloudTrail (audit logging), GuardDuty (threat detection)	CloudTrail logs all API calls — primary source for cloud forensics
Microsoft Azure	Azure VMs, Blob Storage, Azure AD (identity), Azure Monitor, Sentinel (SIEM)	Azure Monitor and Activity Logs provide detailed forensic trails
Google Cloud Platform (GCP)	Compute Engine, Cloud Storage, BigQuery, Cloud Audit Logs, Chronicle (SIEM)	Cloud Audit Logs cover admin activity, data access, system events
Alibaba Cloud	ECS, OSS storage, ActionTrail (audit), Security Center	Growing in Asia-Pacific — ActionTrail provides API audit logging

Q5(c)
[2M]

What is the Honeynet Project?

[2 Marks]

● VM & Network Forensics

The Honeynet Project is an international, non-profit security research organization that develops and deploys honeynets — networks of honeypot systems designed to be attacked — to study attacker tools, tactics, techniques, and procedures (TTPs) in a controlled, monitored environment.

- **Honeynet vs Honeypot:** A honeypot is a single decoy system; a honeynet is a network of multiple honeypots creating a realistic-looking target environment with full network traffic capture capabilities
- **Forensic use:** All traffic to/from honeynets is captured and analyzed using Wireshark, network IDS, and forensic tools — provides real-world malware samples, attacker methodologies, and zero-day exploits
- **Data capture tools:** Sebek (kernel-level data capture), Argus (network flow capture), Snort (IDS), tcpdump
- **Output:** Publishes Know Your Enemy whitepaper series and threat intelligence shared with the security community

■ PREDICTED QUESTIONS — High-Probability Exam Topics

HOW TO USE These predicted questions are based on syllabus topics NOT covered in the previous year paper, common exam patterns for SPPU, and topics marked in the syllabus as high-credit topics. Study these carefully — they are likely to appear.

PREDICTED
1

Explain the Android file system hierarchy and its importance in forensics.

[5 Marks]

■ PREDICTED | ● Mobile Forensics — Ch.2 High Priority

Android File System Overview

Android uses a Linux-based file system with a hierarchical directory structure. Understanding this structure is essential for forensic examiners to know where evidence is located.

Directory	Contents	Forensic Importance
/data/data/	Application private data (SQLite databases, shared prefs, caches) — one subdirectory per installed app	CRITICAL — contacts, messages, app credentials stored here
/data/media/	Internal storage files (photos, downloads) accessible as 'Internal Storage'	User photos, downloaded files, documents
/sdcard/ or /storage/	External storage — photos, videos, documents, downloads	Rich in user-generated content; easy to acquire
/system/	OS files, installed system apps, framework files	Rarely modified; alterations indicate tampering
/proc/	Virtual filesystem with kernel/process info (live system only)	Network info, CPU usage, process details for live forensics
/cache/	Application and system cache data	Can contain remnants of deleted messages and browsing data

Root access is required to access /data/data/ — without root, only /sdcard/ contents are accessible via ADB without special methods. Forensic tools bypass this restriction using physical acquisition or exploiting vulnerabilities.

PREDICTED
2

What is the Cyber Kill Chain? How is it relevant to digital forensics investigations?

[5 Marks]

■ PREDICTED | ● Network Forensics — Linked to CTI syllabus

The Cyber Kill Chain (developed by Lockheed Martin) is a framework that describes the sequential phases of a cyber attack. Understanding these phases helps forensic investigators reconstruct attacker actions from digital evidence.

Phase	Attacker Action	Forensic Evidence to Look For
1. Reconnaissance	OSINT gathering, scanning, target profiling	Web server logs, DNS query logs, port scan logs

2. Weaponization	Create malware payload + exploit delivery vehicle	Malware samples, weaponized documents, exploit code
3. Delivery	Send phishing email, USB drop, drive-by download	Email headers, phishing emails, web proxy logs
4. Exploitation	Execute exploit, gain initial access	Crash dumps, event logs (4624, 4625), IDS alerts
5. Installation	Install backdoor/RAT for persistence	Registry Run keys, scheduled tasks, new services, Prefetch
6. C2 (Command & Control)	Establish communication channel to attacker server	Netflow logs, unusual outbound connections, DNS queries to DGAs
7. Actions on Objectives	Data exfiltration, ransomware encryption, lateral movement	Large outbound transfers, encrypted files, lateral movement logs

PREDICTED
3

Explain the Order of Volatility in digital forensics. Why must it be followed?

[5 Marks]

■ **PREDICTED** | ● **General Forensics — Frequently Examined**

The Order of Volatility (RFC 3227) defines the sequence in which digital evidence must be collected, prioritizing the most volatile data (data that disappears soonest) first to prevent evidence loss during investigation.

Priority	Evidence Type	Retention Duration	Example
1 (Most Volatile)	CPU registers, cache, running processes	Lost immediately on power off	Encryption keys, malware in RAM
2	RAM contents, routing tables, ARP cache	Lost when system powers off	Open connections, running processes
3	Temporary file system (tmpfs)	Lost on reboot	Session tokens, temp files
4	Network state, active connections	Changes rapidly	Open sockets, active downloads
5	Running processes and their memory	Lost on termination/reboot	Injected code in process memory
6	Disk/Storage (non-volatile)	Persists until overwritten	Files, registry, logs
7 (Least Volatile)	Backups, archive media, cloud storage	Long-term persistence	Backup tapes, cloud snapshots

Why follow this order? Forensic investigation actions themselves consume system resources and can overwrite volatile data. If disk forensics is performed first, RAM contents may be flushed. If network state is

not captured immediately, active connections close. Following RFC 3227 Order of Volatility ensures maximum evidence preservation.

PREDICTED
4

Explain EXIF metadata in digital photographs and its role in forensic investigations.

[5 Marks]

■ **PREDICTED** | ● **Multimedia Forensics — High Value Topic**

EXIF (Exchangeable Image File Format) is a standard metadata format embedded in digital photographs (JPEG, TIFF, RAW) by cameras and smartphones at the moment of capture. EXIF data contains rich forensic information about the device, circumstances, and location of a photo.

Key EXIF Fields and Forensic Value

EXIF Field	Data Stored	Forensic Use
GPS Coordinates	Latitude/Longitude/Altitude of capture location	Places suspect at a specific location at a specific time
DateTime Original	Date and time photo was taken (from camera clock)	Establishes timeline; detect tampering by comparing with file system timestamps
Make / Model	Camera/phone manufacturer and model	Links photo to specific device; corroborates device ownership
Software	Editing software used (Photoshop, GIMP, etc.)	Detects photo manipulation or fabrication
Serial Number (MakerNote)	Camera's unique serial number	Definitively links image to a specific physical camera
Thumbnail Image	Embedded thumbnail (may show original before cropping/editing)	Reveals original uncropped/unedited image content

Tools: ExifTool (command-line), Autopsy EXIF module, Jeffrey's Exif Viewer (web-based). **Anti-forensics risk:** EXIF data can be stripped or falsified by editing software. Investigators must verify EXIF against other corroborating evidence (cell tower records, witness accounts).

PREDICTED
5

Explain social media forensics on mobile devices.

[5 Marks]

■ **PREDICTED** | ● **Email & Social Media — Directly in Syllabus**

Social media applications on mobile devices store extensive forensic evidence locally in SQLite databases and cached files. Forensic examination of these artifacts can reconstruct a suspect's communications, location, contacts, and activities.

Evidence Stored by Social Media Apps on Android

- **WhatsApp:** /data/data/com.whatsapp/databases/msgstore.db — all messages, media references; wa.db — contacts list. Media stored in /sdcard/WhatsApp/Media/
- **Facebook:** /data/data/com.facebook.katana/ — posts, messages, contacts, login tokens stored in multiple SQLite databases and shared preference files
- **Instagram:** /data/data/com.instagram.android/ — direct messages, cached profile photos, session tokens
- **Snapchat:** Designed for ephemerality — but cached images may persist in /data/data/com.snapchat.android/cache/ for a period before deletion

Forensic Extraction Process

1. Acquire device image using logical or physical method (ADB backup or physical imaging)
2. Navigate to /data/data/[app.package.name]/ directory in acquired image
3. Extract all .db SQLite database files
4. Open in DB Browser for SQLite or use Autopsy Android Analyzer
5. Parse message tables, contact tables, and media reference tables
6. Recover deleted messages using SQLite freelist page analysis
7. Extract cached images and thumbnails from cache/ directories
8. Map timestamps in SQLite databases to UTC and correlate with timeline

EXAM TIP

When WhatsApp encryption keys are available from the device, encrypted database backups (msgstore.db.crypt14) can be decrypted using the 'key' file from /data/data/com.whatsapp/files/ and tools like WhatsApp Crypt DB Decrypter.

PREDICTED
6

What are image header repairing techniques in multimedia forensics?

[5 Marks]

■ PREDICTED | ● Multimedia Forensics

Digital image files begin with specific file signatures (magic bytes) and structured headers that define format parameters. When files are damaged (corrupted headers), they cannot be opened by standard software — forensic header repair recovers these files.

Common Image File Signatures (Magic Bytes)

Format	Magic Bytes (Hex)	ASCII Representation
JPEG	FF D8 FF	ÿØÿ
PNG	89 50 4E 47 0D 0A 1A 0A	.PNG....
GIF	47 49 46 38 37 61 / 39 61	GIF87a / GIF89a
BMP	42 4D	BM
TIFF	49 49 2A 00 / 4D 4D 00 2A	II* / MM.*

Header Repair Process

1. Open damaged file in WinHex or 010 Editor hex editor
2. Identify which header bytes are missing or corrupted
3. Compare with a known-good sample of the same file format
4. Replace corrupted header bytes with correct values based on format specification
5. Verify file structure integrity (for JPEG: scan for proper SOS/EOI markers; for PNG: validate CRC checksums of each chunk)
6. Test repaired file by opening in image viewer; refine if display artifacts remain
7. Compute hash of repaired file and document all changes made

Tools: WinHex, 010 Editor (with binary templates), JPEG-repair tools, PhotoRec, Autopsy.

QUICK REFERENCE — Key Terms & Definitions

Term	Definition / Key Points
Prefetch Artifacts	C:\Windows\Prefetch\ .pf files; prove program execution; contain: app name, run count, last run time, accessed files
File Carving	Extract files from raw images using file signatures (magic bytes) without file system metadata; tools: Foremost, Scalpel
Steganography	Hiding data inside carrier files without visible change; most common: LSB in images; detected by steganalysis
Steganalysis	Detecting hidden data; methods: visual, statistical (chi-square), signature-based; tools: StegDetect, StegExpose
Logical Extraction	Mobile forensics method using OS APIs (ADB); fast but cannot recover deleted data
Physical Extraction	Bit-by-bit mobile image; can recover deleted data; requires root/exploit; most comprehensive
JTAG / Chip-Off	Hardware-level acquisition for locked/damaged devices; JTAG uses debug port; Chip-Off removes flash chip
SQLite Recovery	Deleted records persist in SQLite freelist pages until reused; tools: undark, sqlparse, Autopsy
Live Acquisition	Evidence collection from running system; captures volatile data (RAM, connections, running processes)
Order of Volatility	RFC 3227: CPU registers → RAM → Temp files → Network state → Disk → Backups
Type 1 Hypervisor	Bare-metal hypervisor on hardware directly; examples: ESXi, Hyper-V, Xen; high performance
Type 2 Hypervisor	Hosted hypervisor on host OS; examples: VMware Workstation, VirtualBox; used in forensic labs
SLA (Cloud)	Contract defining CSP obligations: data access, log retention, breach notification, forensic cooperation
Multi-tenancy	Multiple customers share cloud infrastructure; prevents full disk imaging; requires targeted acquisition
EXIF Metadata	Photo metadata: GPS, timestamp, device make/model, software; tool: ExifTool
Email Header Analysis	Read Received: chain bottom-up for origin IP; check DKIM/SPF/DMARC for authenticity
Honeynet	Network of honeypot systems designed to be attacked; captures attacker TTPs; run by Honeynet Project
Cloud Forensics Evidence	CloudTrail/Audit logs, VM snapshots, VPC flow logs, IAM logs, S3 access logs, local Prefetch files
Lossless Compression	No data loss; perfect reconstruction; examples: PNG, ZIP, TIFF; preferred for forensic evidence
Lossy Compression	Permanent data loss for higher compression; examples: JPEG, MP3, AAC; destroys steganographic content

■ EXAM STRATEGY

Q1: Attempt all 12 — answer any 10. Keep answers to 2–3 lines max. Use definition + example format. **Q2:** Attempt any 5 of 6. Use 4–5 bullet points. ~8–10 lines per answer. **Q3 & Q4:** Attempt any 4 each. Structure with headings. 15–20 lines per answer. **Q5:** One full question with sub-parts [5+3+2]. Plan your time — Q5(a) needs detailed paragraphs. **Always:** Draw diagrams wherever possible (email path diagram, mobile forensics flow, hypervisor architecture). Diagrams earn bonus marks.